FPT ACADEMY INTERNATIONAL

FPT – APTECH COMPUTER EDUCATION

MediFlow

A Microservices – Based Distributed Hospital Management System

Supervisor: *LÊ THANH NHÂN*

Semester: *04*

Batch No: *T2.2410.E0*

Group No: *04*

Order:	**Full name**	**Roll No.**
1.	*Phạm Đăng Vinh*	*Student1614910*
2.	*Lê Quang Huy*	*Student1394600*
3.	*Trần Hoàng Anh*	*Student1614884*
4.	*Thân Thế Lộc*	*Student1571994*

***Month**: 10 **Year**: 2026*

This is to certify that

Mr. PHAM DANG VINH

Mr. LE QUANG HUY

Mr. TRAN HOANG ANH

Mr. THAN THE LOC

Have successfully Designed & Developed

MEDIFLOW

A MICROSERVICES – BASED DISTRIBUTED HOSPITAL MANAGEMENT SYSTEM

Submitted by:

Mr. LE THANH NHAN

Date Of Issue:

Authorized Signature:

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Acknowledge

As we approach the completion of our **MediFlow – Microservices-Based Distributed Hospital Management System** project, we are filled with gratitude, pride, and a sense of accomplishment. This project has been a valuable opportunity to apply our academic knowledge, strengthen our technical skills, and gain practical experience in developing complex software systems.

Heartfelt Gratitude to Our Mentors

First and foremost, we would like to express our sincere gratitude to the **Project Team at FPT Aptech** for providing us with the knowledge, resources, and opportunity to undertake this meaningful project.

We are especially thankful to our lecturer, **Mr. LE THANH NHAN**, for his dedicated guidance, patience, and valuable feedback throughout the development process. His mentorship has helped us overcome technical challenges, improve our problem-solving abilities, and develop a more professional approach to software engineering.

Appreciation for Our Team and Families

We extend our heartfelt appreciation to every member of **Team 04** for their dedication, teamwork, and perseverance. Through close collaboration, we have explored and developed a distributed healthcare platform encompassing ten business domains: **Organization, Patient, Clinical, Laboratory, Pharmacy, Billing, Notification, Reporting, Inpatient, and Surgery**.

We are equally grateful to our families, friends, and classmates for their continuous encouragement and understanding throughout this journey.

A Meaningful Learning Experience

The **MediFlow Project** has provided us with valuable experience in microservices architecture, distributed systems, database design, event-driven communication, and cross-platform application development. By working with technologies such as **Spring Boot, Spring Cloud, RabbitMQ, PostgreSQL, Next.js, and Flutter**, we have gained a deeper understanding of modern software engineering practices and the challenges of integrating independent healthcare services.

Beyond technical knowledge, this project has strengthened our communication, adaptability, responsibility, and collaborative problem-solving skills.

Looking Forward

As we conclude this important milestone, we believe the knowledge and experience gained from MediFlow will serve as a strong foundation for our future careers in information technology.

Once again, we sincerely thank everyone who has supported and contributed to our journey.

With heartfelt gratitude,

TEAM 04

MediFlow – Microservices-Based Distributed Hospital Management System

Synopsis

MediFlow – Microservices-Based Distributed Hospital Management System is a healthcare management project designed to address the fragmentation of patient information, hospital operations, and medical workflows across different departments. The system aims to improve coordination, streamline administrative processes, and support efficient healthcare service delivery through a distributed software architecture.

Built on a **microservices-based and event-driven architecture**, MediFlow encompasses ten business domains: **Organization, Patient, Clinical, Laboratory, Pharmacy, Billing, Notification, Reporting, Inpatient, and Surgery**. These services are designed to manage distinct healthcare responsibilities while enabling coordinated workflows, including patient registration, appointment scheduling, clinical examinations, laboratory testing, prescription management, payment processing, medication dispensing, inpatient care, and surgical procedures.

The project utilizes **Java Spring Boot, Spring Cloud Gateway, Netflix Eureka, RabbitMQ, and PostgreSQL** for backend development, service discovery, asynchronous communication, and data management. **Next.js and Flutter** are used to develop web and mobile applications, providing role-based interfaces for healthcare professionals and administrative personnel.

MediFlow follows essential software engineering principles, including **Domain-Driven Design, Clean Architecture, database-per-service ownership, RESTful API communication, and event-driven integration**. These approaches promote modularity, maintainability, scalability, and separation of responsibilities across independent services.

As an academic project under active development, MediFlow demonstrates the practical application of distributed systems and modern software engineering technologies in the healthcare domain. Beyond its functional scope, the project provides valuable experience in system architecture, collaborative development, integration testing, and solving complex real-world business problems.

Keywords: Hospital Management System, Healthcare Information System, Microservices, Distributed Systems, Event-Driven Architecture, Spring Boot, RabbitMQ, Next.js, Flutter.

Review 01

Problem Definition

1. Executive Context

Hospitals coordinate information that is simultaneously personal, clinical, operational, and financial. A patient may interact with reception, a doctor, nursing staff, a laboratory, a pharmacy, a cashier, an inpatient ward, and a surgery team during a single episode of care. Each department creates information needed by another department, yet each remains responsible for its own professional decisions and records.

When this information is managed by paper, spreadsheets, isolated applications, or informal messages, the organization loses a reliable end-to-end view. Staff repeat data entry, status changes are communicated late, charges may be missed or duplicated, inventory can become inaccurate, and management reports depend on manual reconciliation. Sensitive data is also harder to protect because access rules are inconsistent across tools.

MediFlow is designed as a hospital workflow platform that connects these areas while preserving clear data ownership. The current repository implements a browser-based frontend and a Spring Boot microservice backend. The final system comprises ten business services supported by an API gateway, Eureka service discovery, PostgreSQL databases, and RabbitMQ messaging.

The central design principle is that each business service is authoritative for its own data. Other services exchange UUID references, use REST only when immediate authoritative information is required, and use events to distribute completed business facts. This structure supports traceability and reduces hidden cross-department coupling.

2. Problem Identification

Problem	Existing operational impact	Required system response
Fragmented patient information	Staff cannot obtain a complete and timely view of a patient	Maintain a consistent patient identity and connect authorized records by UUID
Manual appointment coordination	Double booking, missed appointments, and slow check-in can occur	Provide appointment registration, status tracking, and encounter handoff
Disconnected clinical documentation	Diagnoses, observations, and orders may not reach the next department reliably	Record clinical encounters and publish controlled downstream requests
Slow laboratory turnaround visibility	Clinicians may not know whether a test is ordered, collected, processed, or completed	Track the laboratory order lifecycle and publish verified results
Prescription and stock separation	Medicines may be prescribed without clear dispensing or inventory status	Authorize prescriptions, validate stock, dispense medicine, and record movements
Inconsistent financial tracking	Charges, invoices, payments, deposits, receipts, and refunds may be reconciled manually	Centralize financial records in the Billing context and expose auditable status
Weak inpatient coordination	Admission, financial clearance, bed allocation, care	Provide one stateful inpatient workflow with explicit transitions

Problem	Existing operational impact	Required system response
	stay, and discharge may be tracked separately	
Weak surgery coordination	Readiness, clearance, scheduling, execution, and outcomes may be communicated informally	Provide a traceable surgery lifecycle with authorization checkpoints
Delayed communication	Patients and staff may not receive important status changes promptly	Generate notifications from domain events and retain delivery status
Limited management visibility	Managers may depend on manually assembled reports	Build reporting read models from trusted operational events
Insufficient access control	Sensitive medical and financial data may be exposed to unauthorized users	Enforce role-based access at every backend endpoint
Poor auditability	It may be difficult to determine who changed a record and when	Retain identifiers, timestamps, statuses, and event histories suitable for audit

3. Project Objectives

The primary objective is to provide a coherent, secure, and traceable digital workflow for a patient's journey across hospital departments.

Specific objectives are:

- Establish traceable end-to-end workflows from registration to care, diagnostics, medication, billing, inpatient care, surgery, notification, and reporting.
- Give each bounded context clear ownership of its data and business rules.
- Reduce duplicate entry by exchanging identifiers and authorized information through explicit contracts.
- Enforce role-based authorization at the gateway-facing backend endpoints.
- Record state changes and publish domain events for dependent workflows.
- Support complete financial accountability through charges, invoices, deposits, payments, receipts, refunds, settlement, and service clearance.
- Improve coordination across appointments, encounters, Laboratory, Pharmacy, Billing, Inpatient, and Surgery workflows.
- Provide reliable background processing through outbox publication, idempotent consumers, retry handling, and dead-letter queues.
- Supply managers with consolidated operational information without allowing reporting logic to modify source-of-truth records.
- Maintain a modular architecture that can be developed, tested, and deployed by bounded context.

4. Project Scope

4.1. Included Scope

Scope area	Included capabilities
Organization	Users, staff, departments, roles, facilities, rooms, and organization reference data
Patient	Patient profiles, identifiers, contact information, and demographic records
Clinical	Appointments, encounters, medical records, diagnoses, and admission-referral references; examination requires matching EXAM clearance or an audited emergency override
Laboratory	Tests accepted from explicit Clinical requests, specimens, processing states, results, and publication; protected processing requires matching LAB_TEST clearance or an audited emergency override
Pharmacy	Medicine catalogue, Pharmacy-owned prescription creation, inventory, payment-proof validation, stock movement, and dispensing
Billing	Charges, invoices, deposits, payments, receipts, refunds, settlement, and purpose-specific service clearance
Notification	Event-driven notification requests, templates, delivery attempts, and delivery status
Report	Event-fed read models and management-oriented operational reports
Inpatient	Admission requests, financial clearance, bed ownership/allocation, active stays, treatment, transfers, medical discharge, settlement, and administrative close

Scope area	Included capabilities
Surgery	Surgery requests, readiness review, financial clearance, scheduling, execution, recovery, outcomes, and charge handoff
Shared platform	API gateway, service discovery, per-service databases, messaging, security conventions, and shared contracts
Web client	Role-aware pages that call the platform through same-origin /api/* routes

4.2. Excluded or not yet proven complete

- Production integration with national identity, health insurance, banking, payment gateways, or government systems.
- Certified interoperability with external laboratories, medical devices, PACS, or third-party electronic health records.
- Production delivery contracts with SMS, email, or push-notification vendors.
- A complete runnable Flutter mobile application; the repository currently contains only a shared loader package.
- Clinical decision support, automated diagnosis, prescription recommendation, or AI-assisted treatment decisions.
- A finalized production service-level objective, retention policy, disaster-recovery target, or clinical governance process.
- Formal regulatory certification, accessibility certification, penetration-test certification, or medical-device certification.

5. Users and Functions by Role

Role	Main responsibilities	Representative system functions
ADMIN	Configure and govern the platform	Manage users, roles, departments, facilities, organization reference data, and permitted Inpatient bed records; review administrative status
DOCTOR	Diagnose and direct care	Review patient information; update encounters; enter observations/diagnoses; request tests; prescribe; create admission and surgery requests; follow care outcomes
NURSE	Coordinate permitted patient and ward workflows	Create permitted patient/appointment records; support appointment/check-in/read operations; record permitted Inpatient treatment observations; perform authorized bed operations
PHARMACIST	Validate and dispense medication	Review authorized prescriptions; verify medicines and quantities; manage stock; record dispensing and inventory movement
CASHIER	Process authorized financial transactions	Create and settle invoices; receive deposits and payments; issue receipts; process authorized refunds; confirm settlement and service clearance
LAB_TECH	Process existing diagnostic tests	Start/cancel permitted tests; manage specimens; record status/results; verify and

Role	Main responsibilities	Representative system functions
		publish results; test creation follows an authorized Clinical request and belongs to Laboratory
MANAGER	Monitor hospital operations	View operational and financial reports, service activity, workloads, trends, and management indicators
PATIENT	Access personal healthcare information and services	Manage permitted profile data; request and review appointments; view authorized results, prescriptions, invoices, receipts, payments, and notifications
SYSTEM	Execute trusted automation	Publish and consume events; retry failed delivery; update read models; initiate notifications; preserve idempotency and traceability

6. Proposed Solutions

MediFlow proposes a modular hospital platform in which each business capability is implemented as an independent Spring Boot service. The services are accessed through one gateway and registered through Eureka. Each service owns a separate PostgreSQL database, preventing direct cross-service database access. When one service requires authoritative information from another, it uses a defined REST contract. When a completed state change must inform multiple downstream consumers, it publishes an event through RabbitMQ.

Solution element	Purpose	Expected benefit
Next.js web client	Provide role-aware workflows in a consistent user interface	Faster staff access and fewer disconnected screens
API gateway	Expose one controlled entry point	Consistent routing, authentication context, and reduced client coupling
Independent services	Separate business ownership and deployment	Clear boundaries, safer change, and focused testing
Isolated PostgreSQL databases	Keep each service authoritative for its records	Reduced accidental coupling and clearer accountability
REST contracts	Resolve information required immediately for a command	Timely validation using the owning service

Solution element	Purpose	Expected benefit
RabbitMQ events	Propagate completed business facts	Decoupled notifications, reporting, and cross-department reactions
Transactional outbox	Coordinate database changes with event publication	Lower risk of losing events after a committed state change
Idempotent consumers	Prevent repeated delivery from creating repeated business effects	Safer retries and recovery
Role-based authorization	Restrict every protected operation	Better privacy and separation of duties
Automated tests	Verify rules, contracts, persistence, and integrations	Repeatable evidence that implemented behavior remains stable

Customer Requirements Specifications

1. Stakeholders and Customer Requirements

Stakeholder	Core requirement	Success indicator
Hospital administration	Govern users, departments, facilities, and access	Authorized configuration is available and changes are traceable
Clinical staff	See relevant patient information and document care efficiently	Encounters, orders, diagnoses, and care states are available to permitted roles
Laboratory staff	Receive complete orders and publish verified results	Each order has an identifiable lifecycle from request to published result
Pharmacy staff	Dispense only valid prescriptions with sufficient stock	Dispensing creates auditable medication and inventory records

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Stakeholder	Core requirement	Success indicator
Finance staff	Reconcile the complete patient financial lifecycle	Charges, invoices, deposits, payments, receipts, refunds, settlement, and clearance have auditable states
Inpatient staff	Coordinate admission, clearance, bed use, stay, and discharge	No admission proceeds without required state and resource checks
Surgery team	Coordinate readiness, clearance, scheduling, execution, recovery, and outcomes	Every case progresses through an authorized, auditable lifecycle and produces the required care and charge outputs
Managers	Obtain timely operational and financial insight	Reports use trusted source events and do not modify operational data
Patients	Access permitted personal information and services	Authorized self-service exposes only the patient's own profile, appointments, results, prescriptions, invoices, receipts, payments, and notifications

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Stakeholder	Core requirement	Success indicator
Technical operations	Operate services reliably and recover from transient failures	Health, logs, retries, dead-letter handling, and database isolation are observable

2. Data-Processing Cycle by User Role

Workflow	Actor	Input information	Processing	Output information	Owning service
User and role administration	ADMIN	User identity, staff details, assigned roles, department	Validate uniqueness and permissions; create or update organization records	User/staff record, role assignment, audit timestamps	Organization
Organization setup	ADMIN	Facility, department, room, and organization attributes	Validate hierarchy and persist organization reference data	Available organization and room references	Organization

Workflow	Actor	Input information	Processing	Output information	Owning service
Bed catalogue setup	ADMIN, MANAGER	Department/room references and bed attributes	Validate references and bed state; persist the Inpatient-owned resource	Bed catalogue and availability state	Inpatient
Patient registration	ADMIN, NURSE	Demographics, contact data, identifiers	Validate required fields and duplicate constraints; assign UUID	Patient profile and registration confirmation	Patient
Appointment scheduling	ADMIN, NURSE	Patient UUID, department/provider, requested time, reason	Validate request and scheduling rules; create appointment state	Appointment identifier, schedule, and status	Clinical

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Workflow	Actor	Input information	Processing	Output information	Owning service
Appointment/check-in support	ADMIN, DOCTOR, NURSE as permitted by route	Appointment and status input	Apply permitted appointment state; ARRIVED alone does not authorize examination	Updated appointment state	Clinical
Clinical examination and record	ADMIN, DOCTOR	Patient, appointment, findings, matching EXAM clearance or audited emergency override	Validate authority, encounter state, and proof; create/update record and diagnosis	Medical record, diagnosis, disposition/reference	Clinical
Clinical test request and	ADMIN, DOCTOR	Patient/record references, department, test code, requested date, order ID	Validate the clinical request; transfer the explicit order;	Accepted Laboratory test in initial state	Laboratory

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Workflow	Actor	Input information	Processing	Output information	Owning service
Laboratory creation			create the Laboratory-owned test		
Laboratory processing and result	ADMIN, LAB_TECH	Existing Lab test, matching LAB_TEST clearance or audited emergency override, specimen/results	Validate proof and lifecycle; start, record, complete, and publish	Published Laboratory result	Laboratory
Prescription creation	ADMIN, DOCTOR	Clinical record reference, patient, medicine, dose, quantity, instructions	Pharmacy validates lines/prices and creates the Pharmacy-owned prescription plus pending dispense slip	Prescription and medication lines	Pharmacy

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Workflow	Actor	Input information	Processing	Output information	Owning service
Medicine dispensing	ADMIN, PHARMACIST	Pharmacy prescription, durable payment proof, stock, dispense quantity	Verify proof/prescription/stock; record dispense and inventory movement	Dispense record and remaining stock	Pharmacy
Invoice and payment processing	Authorized Billing staff	Patient/source context, charge, invoice, and payment data	Validate financial state; create invoice; record payment and receipt	Invoice, payment state, and receipt	Billing
Deposit, refund, settlement, and clearance	ADMIN, CASHIER	Payment request, original transaction, service purpose, and admission/care reference	Apply append-only and idempotent financial rules; authorize refund or service clearance	Deposit, refund, settlement, and clearance records	Billing

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Workflow	Actor	Input information	Processing	Output information	Owning service
Inpatient admission creation	ADMIN, DOCTOR	Patient, clinical need, and admission details	Validate references and create the admission	Pending admission	Inpatient
Bed assignment and transfer	ADMIN, NURSE	Admission and requested/target bed	Validate availability and perform the bed transition	Updated admission and bed occupancy	Inpatient
Inpatient treatment	ADMIN, DOCTOR, NURSE	Active admission and treatment/observation information	Append or correct permitted treatment entries	Updated treatment history	Inpatient
Explicit inpatient admit	ADMIN, DOCTOR, NURSE	Pending admission, active bed assignment, matching deposit clearance or audited emergency override	Invoke the permitted /admit transition after both normal gates	ADMITTED state	Inpatient

Workflow	Actor	Input information	Processing	Output information	Owning service
Medical discharge	ADMIN, DOCTOR	Active admission and clinical discharge information	Apply medical-discharge transition	Medically discharged admission	Inpatient
Administrative close	ADMIN, CASHIER	Medically discharged admission plus matching settlement.completed or approved debt/emergency override	Validate the close gate and apply administrative close	Closed admission and released resources	Inpatient
Surgery lifecycle	Authorized surgery staff	Request, patient/clinical context, readiness, clearance, schedule, team, procedure, recovery, and outcome data	Validate authority and prerequisites; schedule, execute, record recovery/outcome, and publish charge handoff	Auditable surgery case, outcome, and charge event	Surgery

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Workflow	Actor	Input information	Processing	Output information	Owning service
Notification delivery	SYSTEM	Domain event, recipient reference, template data, channel	Select template; create delivery request; attempt and track delivery	Notification content, delivery status, failure details	Notification
Management reporting	MANAGER, SYSTEM	Trusted domain events and report criteria	Update read models; aggregate permitted measures; filter by criteria	Operational and financial report output	Report

3. Cross-Department Information Flows

Flow	Trigger	Main processing path	Result
Registration to appointment	Authorized staff register a patient and schedule care	Patient stores identity; Clinical owns and stores the appointment using the patient UUID	Confirmed appointment associated with the correct patient UUID
Appointment to encounter	Appointment reaches check-in/consultation state	Clinical validates the patient/appointment reference and opens an encounter	Traceable clinical encounter
Clinical decision to Laboratory test	ADMIN/DOCTOR decides a test is needed	Clinical sends an explicit test code and producer-generated order ID; Laboratory validates and creates its test	Laboratory-owned test, later processed only with required clearance/override
Laboratory to billing	A billable test reaches the contractually defined state	Laboratory event is consumed idempotently by Billing	One charge for the eligible laboratory activity

Flow	Trigger	Main processing path	Result
Prescription to dispensing	Doctor creates a Pharmacy-owned prescription	Pharmacy later requires durable payment proof and valid stock before dispensing	Medication supplied with inventory movement history
Care to finance	An approved billable care event reaches Billing	Billing records the charge idempotently, updates the account, and returns the required settlement or service-clearance state	Traceable charge, invoice, payment, and clearance result
Admission to active stay	Admission, active bed assignment, and matching deposit clearance or audited emergency override are present	Authorized role invokes the explicit Inpatient admit transition	ADMITTED stay with bed occupancy
Active stay to discharge	Medical discharge plus matching settlement	Authorized roles apply medical discharge, administrative close, and bed release	Closed local admission and reusable bed

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Flow	Trigger	Main processing path	Result
	completion or approved close override is present		
Surgery lifecycle	Clinical readiness and financial clearance are approved	Surgery schedules the case, records execution and recovery, publishes the outcome, and sends eligible charges to Billing	Completed or cancelled surgery case with auditable outcome and financial handoff
Domain event to notification/report	An eligible state change is committed	Outbox publishes event; Notification and Report consume idempotently	Delivery record and updated reporting read model

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4. Hardware / Software requirements

4.1. Client requirements

4.1.1. Hardware

Client profile	Minimum	Recommended
Staff desktop/laptop	64-bit dual-core CPU, 4 GB RAM, 1366×768 display, stable network	Modern 4-core CPU, 8 GB RAM, Full HD display, wired or reliable Wi-Fi
Patient browser device	Recent smartphone, tablet, or computer with at least 2 GB available memory	Device receiving current OS/browser security updates, 4 GB or more memory
Development workstation	64-bit 4-core CPU, 16 GB RAM, 30 GB free storage	8-core CPU, 32 GB RAM, SSD with 60 GB or more free storage
Shared reception/lab terminal	Keyboard, pointing device, 1366×768 display, reliable network	Full HD display, barcode-capable peripherals where locally required, UPS-backed power

4.1.2. Software

Item	Requirement
Browser	Current Chrome, Edge, Firefox, or Safari with JavaScript, cookies, and HTTPS enabled
Operating system	A vendor-supported Windows, macOS, Linux, Android, or iOS release
Network	Stable access to the deployed HTTPS endpoint through approved firewall/proxy rules
Document tools	Microsoft Word, LibreOffice Writer, or compatible tool for report conversion
Mobile status	Responsive browser access is assumed; a complete native mobile client is not currently delivered

4.2. Server requirements

4.2.1. Hardware

Environment	CPU	Memory	Storage	Intended use
Developer demonstration	8 logical cores	16 GB RAM	40 GB free SSD	Local containers, selected services, frontend, and tests
Team integration	16 logical cores	32 GB RAM	150 GB SSD	Concurrent services, shared testing, sample data, and monitoring
Initial production baseline	24+ logical cores distributed across hosts	64+ GB RAM distributed	500+ GB resilient SSD plus backups	Starting point only; must be validated by performance and recovery tests

4.2.2. Software

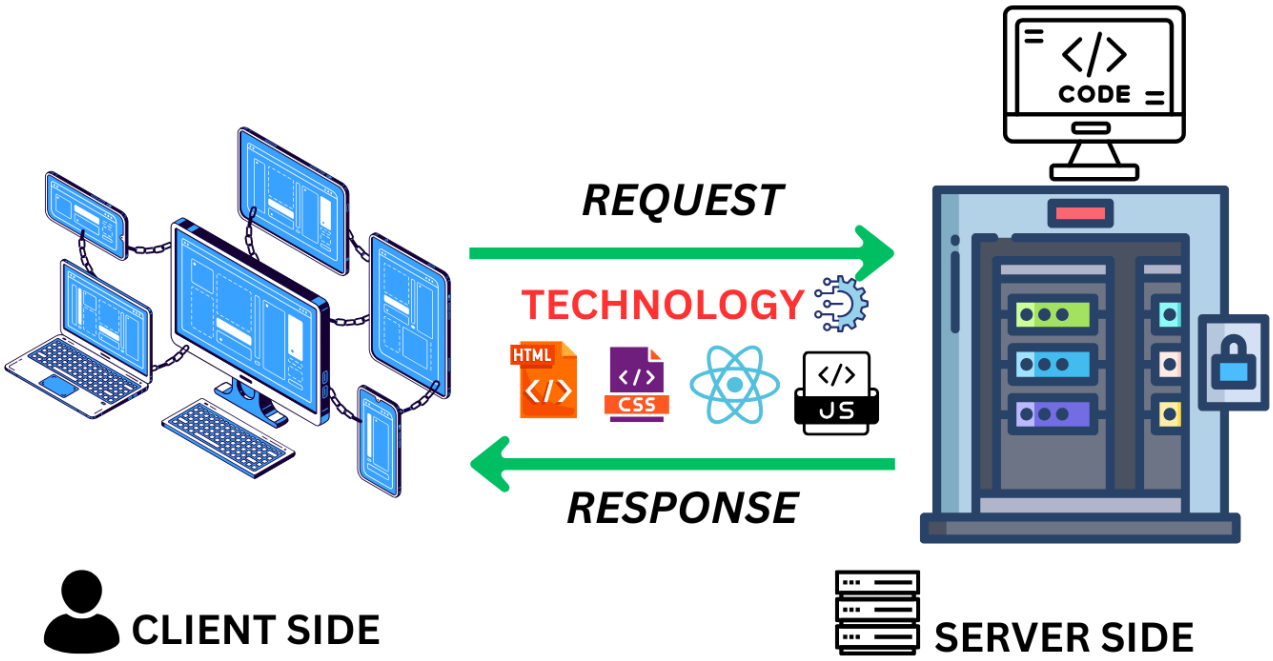
Item	Source-aligned baseline
Java	21
Spring Boot	3.3.5
Spring Cloud	2023.0.3
Backend build	Maven multi-module
Frontend	Next.js 16.2.11, React 19.2.4, TypeScript 5, Tailwind CSS 4, pnpm
Database	PostgreSQL 16-compatible, isolated logical database per service
Messaging	RabbitMQ 3-compatible with operational management support
Discovery	Eureka Server
Containers	Docker Engine and Docker Compose compatible with repository definitions

Item	**Source-aligned baseline**
Production platform services	HTTPS certificate management, secret management, centralized logs/metrics/alerts, backups, time synchronization

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Review 02

Architecture Design



Data Flow Diagram

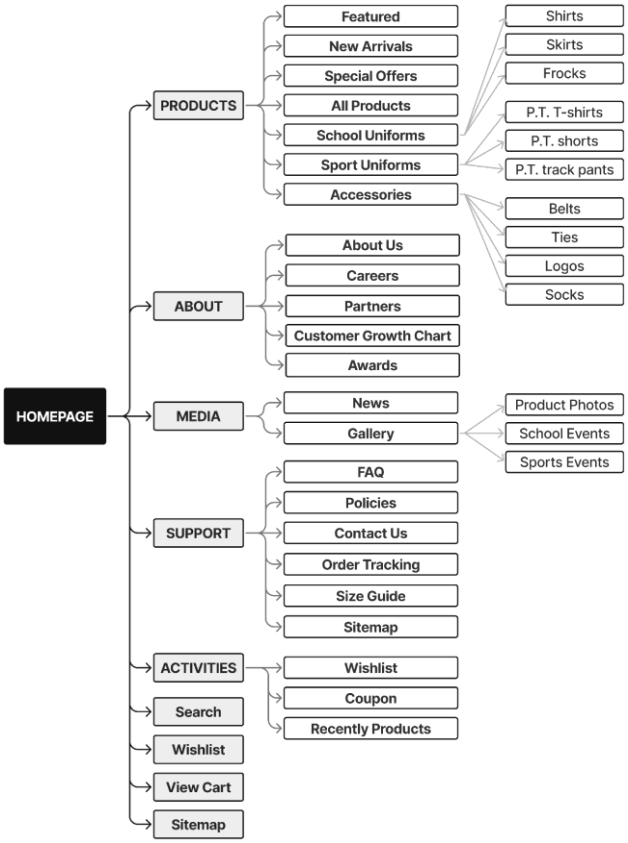
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Entity Relationship Diagram

Table Design

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SiteMap



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Algorithm

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Review 03

Screen shots

1. User Interface Introduction

Before diving into the **Website Structure Overview**, this section provides an overview of the **visual identity, design principles, and tools** used in the **Maverick Dresses** project. These elements ensure consistency, brand recognition, and an optimized user experience across all pages.

2. Brand Identity & Design Concept

2.1. Brand Recognition

Maverick Dresses is a **leading uniform manufacturer**, and the website is designed to reflect its **professional, modern, and trustworthy** identity. The UI/UX approach follows a **minimalist yet elegant** aesthetic, ensuring clarity and ease of navigation for users.

2.2. Color Palette



The color scheme is carefully chosen to balance **brand representation, readability, and visual appeal**. The primary and secondary colors used in the interface include:

- **Primary Colors**:
  - **#f54b49** (Main Accent - Product labels, price highlights)
  - **#eebf8f, #fbece4, #f6eedc** (Supporting Accent - Banners)
  - **#fef2ef** (Top-nav Background & Light Theme Areas)
- **Neutral & Supporting Colors**:
  - **#121212** (Text & Primary Dark UI Elements or Call-to-action button backgrounds)
  - **#454545** (Secondary Text, Menu, Footer)
  - **#7b7b7b** (Disabled Elements, Tertiary Text)
  - **#bababa** (Borders & Subtle Dividers)
  - **#f7f7f7** (Card Backgrounds, Light Mode Sections)

These colors ensure **strong contrast**, **readability**, and **brand consistency** across different devices and screen sizes.

3. Typography & Iconography

3.1. Font Selection

- **Primary Font**: **Jost** (Modern, Clean, and Readable)

This font provides **aesthetic consistency** while maintaining **high readability across desktop and mobile interfaces**.

3.2. Icon Set

The website uses **Lucide Icons & Svg Icon** for a **modern and scalable** icon set. Icons are used for:

- **Navigation Menu & Categories**
- **Cart, Wishlist, and Checkout Steps**
- **Social Media & Interactive Features**
- **Product Status (In Stock, Out of Stock, Discounts, etc.)**

All icons are used in **SVG format** to ensure scalability and performance optimization.

3.3. UI Components & Design Principles

3.1. Navigation & Layout Structure

- **Fixed Header, Scroll to Top button & Tiker** to improve accessibility.
- **Top Navigation Bar (Top-Nav) Includes**:
  - Contact Information (Phone, Email).
  - Free Shipping Information.
- **Primary Navigation Includes**:
  - Products, About, Media, Support, Activities.
  - Search Bar, Cart, and Vistor counter Shortcuts.

3.2. UI Components & Interactions

- **Smooth Hover Effects** on buttons and menus.
- **Interactive Filters & Sorting** on product pages.
- **Collapsible Sections** in FAQ and Policies.
- **Animated Loading Elements** for better user engagement.

3.3. Mobile & Responsive Design

- **Fully Responsive Grid System** (Using **CSS Flexbox & Grid Layouts**).

- **Optimized Navigation Drawer** for mobile users.
- **Touch-friendly Buttons & UI Elements** for smooth user interactions.

3.4. Design Tools & Prototyping

4.1. Design & Prototyping Tool

The entire UI/UX design process was carried out in **Figma**, ensuring a streamlined workflow between **designers and developers**. The Figma file includes:

- **Wireframes & High-Fidelity Mockups**.
- **Interactive Prototypes for Testing**.
- **Component Libraries for Consistency**.
- **Style Guide & Design Tokens**.

4.2. Shared Figma Link

Figma Design File:

MediFlow project evidence: source repository, requirements, architecture, event, and progress documents.

(Accessible for developers, stakeholders, and UI/UX teams).

Important Note:

The Figma file serves as a reference only. During development, there may be **deviations** from the original design due to technical constraints, optimization decisions, or responsive adjustments. The final coded version **may not be 100% identical** to the Figma design.

4. Website Structure Overview: Header, Footer, and Layout Sections

Before diving into the detailed layout of each page, it's important to first define and understand the **Header** and **Footer** components, as they are crucial parts of the website's structure. These sections ensure **consistent branding, seamless navigation, and accessibility** across different pages.

- We strive to optimize the UI/UX as much as possible, ensuring a smooth, intuitive, and enjoyable user experience.

4.1. Header Section



The **Header** is the topmost part of the website and plays a critical role in user interaction. It serves as the **first touchpoint** for visitors, providing key navigation and interactive elements.

Top Navigation Bar (Top-Nav)

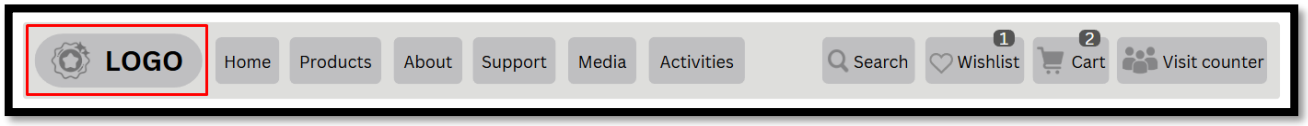


- **Positioned at the very top of the website, above the main navigation menu.**

- **Left Side:** Displays **company contact information**, including:
  - **Phone Number**
  - **Email Address**

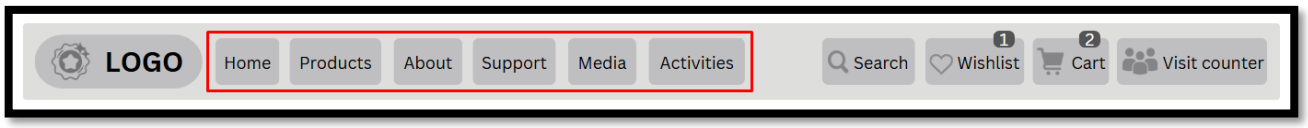
- **Right Side:** Displays **free delivery conditions**, indicating the **minimum purchase amount required** to qualify for free shipping.

Company Logo



- Positioned at the **top-left** of the page.
- Clickable and redirects users to the **Homepage** when clicked.

Main Navigation Menu & Development Plan



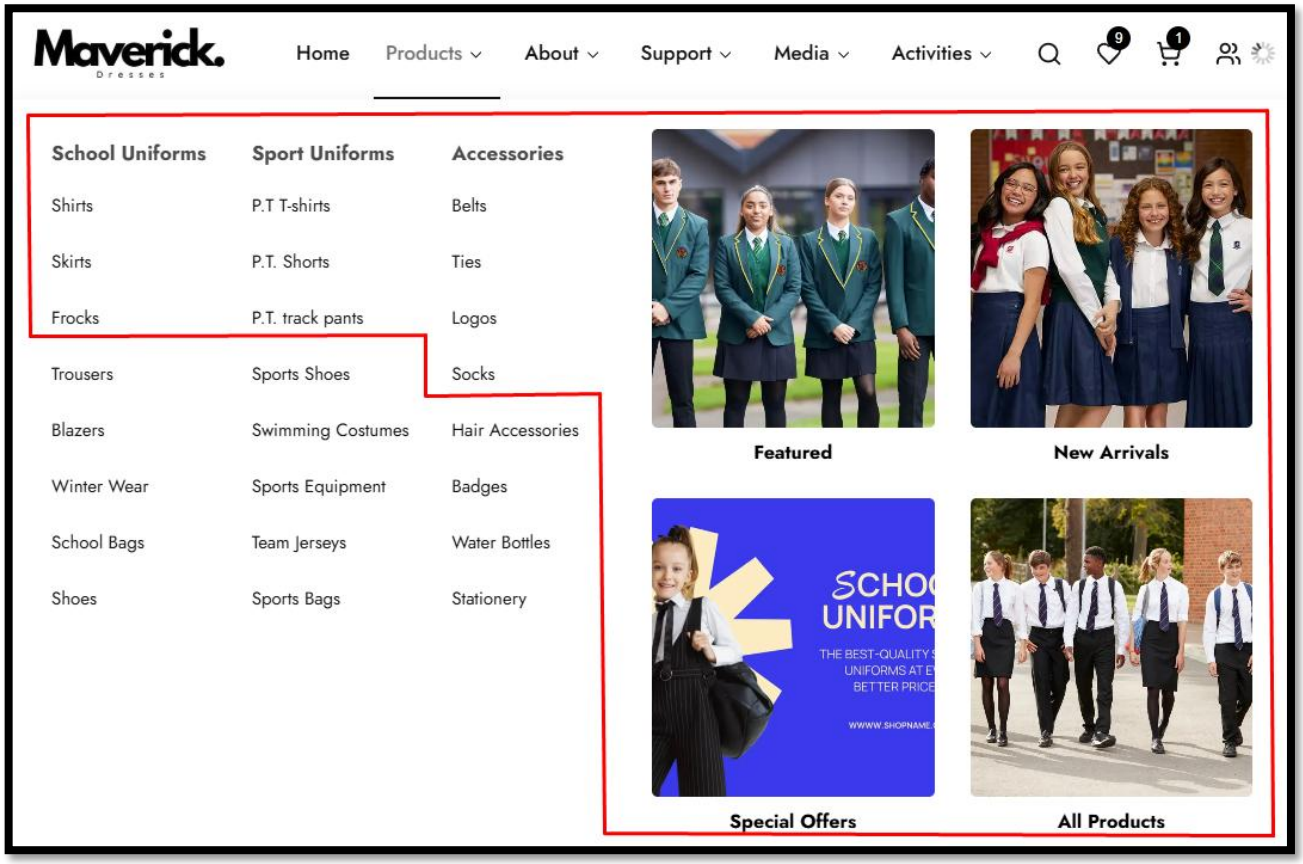
- The **Main Navigation Menu** is positioned at the **top-left** of the page and serves as the primary method for users to navigate through the Maverick Dresses website efficiently. It features a **horizontal** or **vertical menu bar** that provides quick access to major sections.

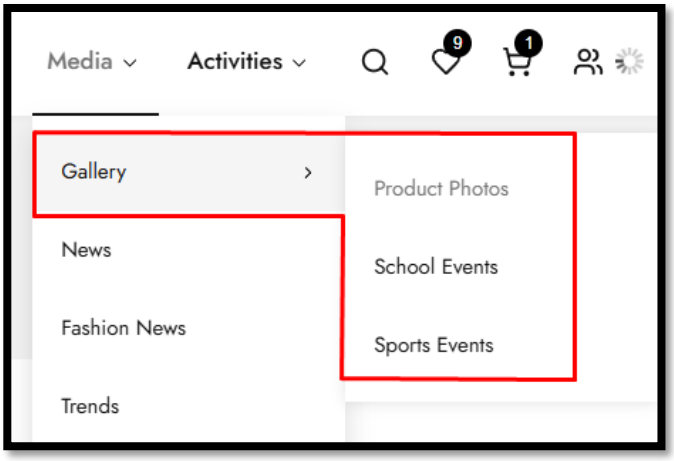
o A horizontal or vertical menu bar that allows users to quickly access major sections:

- **Products** (Featured, New, Sale, School Uniforms, Sports Uniforms, Accessories)
- **About** (About Us, Careers, Partners, Customer Growth Chart, Awards)
- **Media** (Fashion News, Gallery)
- **Support** (FAQ, Policies, Feedback, Contact Us)
- **Activities** (Wishlist, Coupons, Order Tracking)

- **Development Plan (Products & Media Navigation):** To ensure a structured and optimized development process, the Products and Media sections in the Main Navigation Menu are being implemented in two phases.

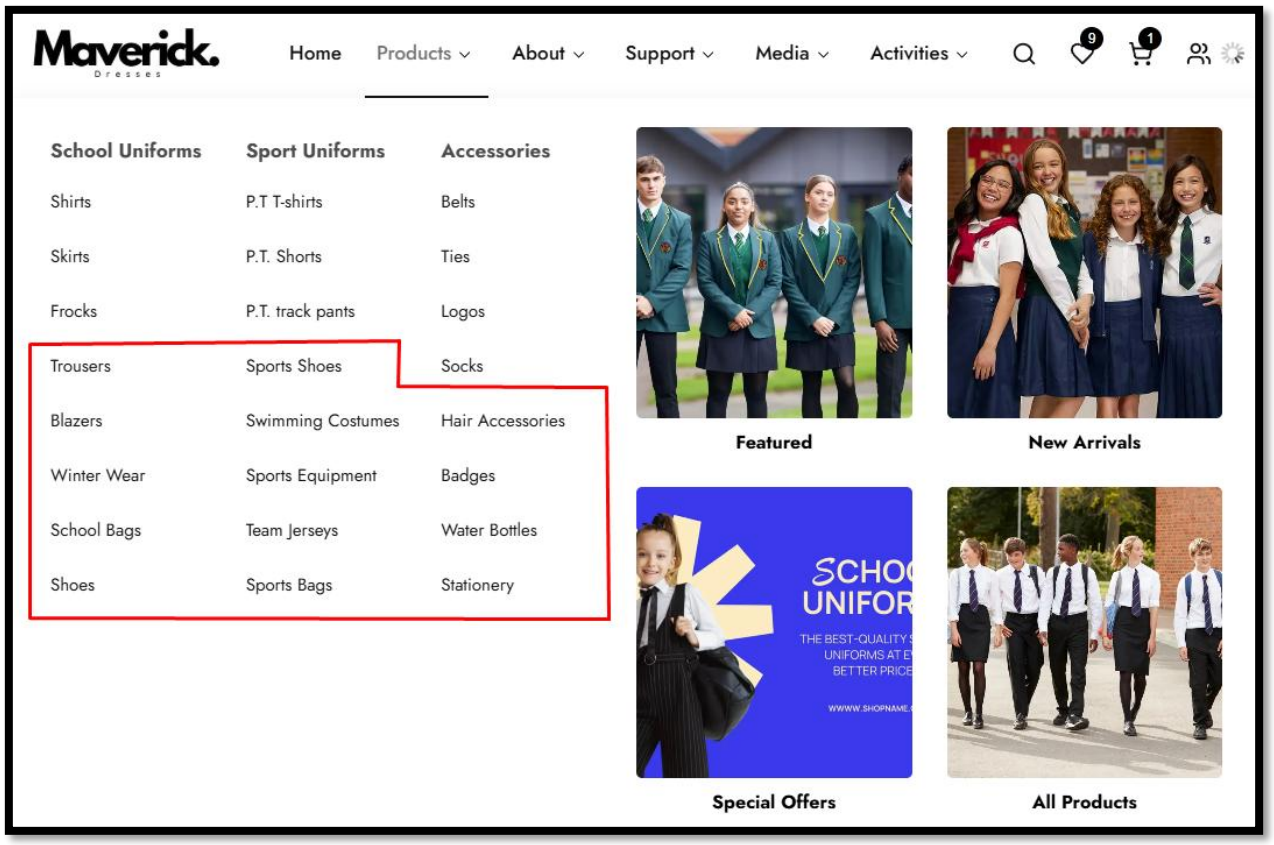
  - **Version 1 - Completed Features**

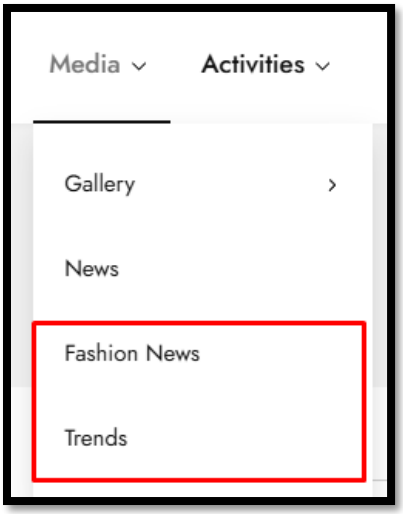




- In **Version 1**, the following **Products** categories and **Media** sections have been fully implemented, with all links functional:
- **Products** -> Featured, New Arrivals, Special Offers, All Products, School Uniforms (Shirts, Skirts, Frocks), Sports Uniforms (P.T. T-Shirts, P.T. Shorts, P.T. Track Pants), Accessories (Belts, Ties, Logos, Socks).
- **Media** -> Gallery (Product Photos, School Events, Sports Events), News

- **Version 2 - Upcoming Enhancements (Currently Not Clickable)**





- The following **sections** are planned for **Version 2**, and their links are **not yet active**:

- **Products** -> Advanced Filtering & Sorting Features for better user experience. Expanded Accessories Section with additional product lines. Detailed Subcategories for School Uniforms & Sports Uniforms, including: (More variations for Shirts, Skirts, and Frocks. Additional uniform styles for different school levels. More Sportswear categories such as custom jerseys, sweatpants, and training gear.).
- **Media** -> Fashion News and Trends – A dedicated section covering worldwide industry updates, style guides, and the latest collections from Maverick Dresses.

- **Version 2 - Upcoming Enhancements (Currently Not Clickable)**
- **By dividing development into two phases, Maverick Dresses ensures:**

- A functional and navigable website in Version 1.
- A scalable and enriched user experience in Version 2.
- A seamless transition between current and future updates.
- With **Version 1 successfully implemented**, the upcoming enhancements in **Version 2** will further elevate the website's **product navigation and media content**, making it a **fully immersive shopping and branding platform**.
- Stay tuned for Version 2!

- **Navbar Highlight for Active Page:** The **Navigation Bar (Navbar)** helps users easily navigate between different sections of the Maverick Dresses website. To improve **user experience and usability**, the active page should be **highlighted** dynamically, allowing users to see which section they are currently visiting.

- **Active Page Indication:**

- The currently selected **menu item** should have a **distinct color change** and/or an **underline** to indicate the active state.
- Example: If a user is on the **Products Page**, the "Products" menu item should be visually

highlighted.

- **Hover Effects:**

- When users **hover over menu items**, a **soft color change** or **underline animation** should appear to indicate interactivity.

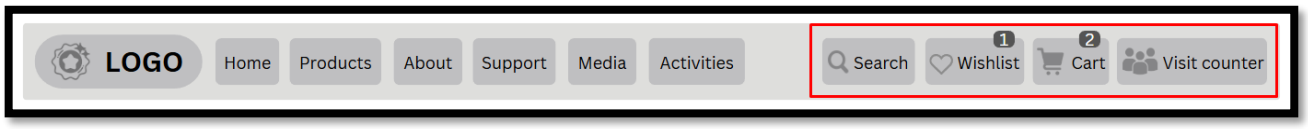
- **Dropdown Menu Active State:**

- If a user selects a **sub-menu item**, both the **main menu and the selected sub-menu item** should be highlighted.
- Example: If a user is on the **FAQ Page**, the "Support" menu (parent category) should also be highlighted.

- **Smooth Transitions:**

- Implement a **fade-in or slide-in animation** when users switch between pages to enhance UX.

Right Icons Menu



Search Bar Icon: Click to show Search bar.

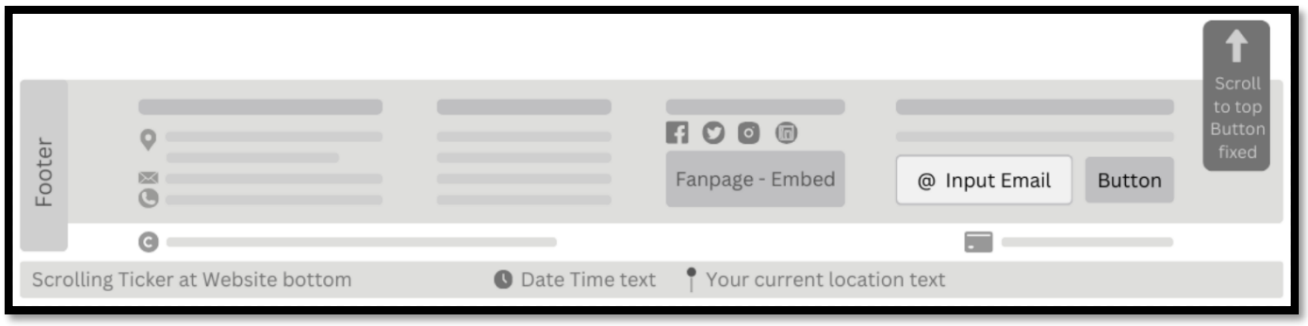
Wishlist Icon: Displays the number of items in the Wishlist.

Cart Icon: Displays the number of items in the cart.

Visitor Counter

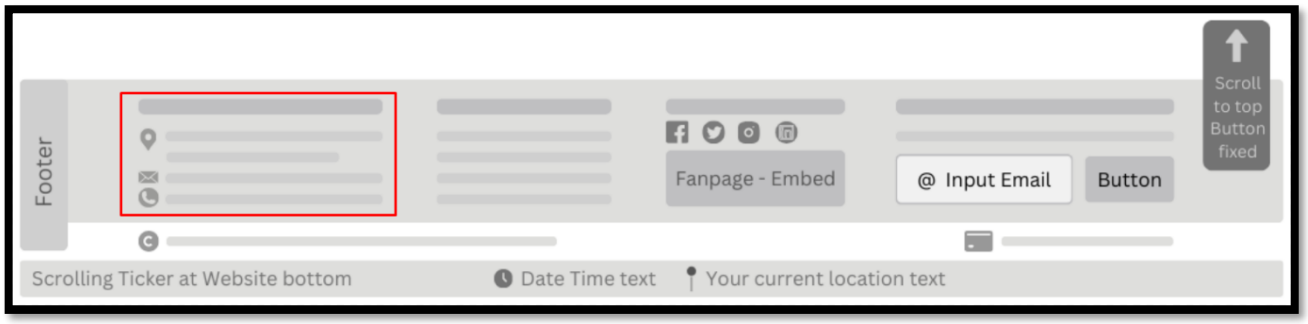
- Positioned near the **logo or within the header section**.
- A **real-time counter** displaying the number of active visitors on the site, creating a **sense of community and engagement**.

4.2. Footer Section



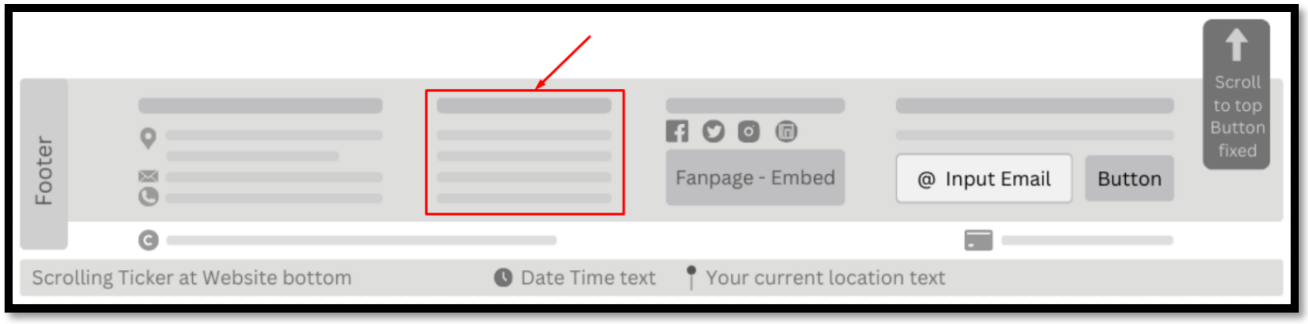
The **Footer** appears at the bottom of each page and provides additional resources for users, such as company details, policies, and quick navigation.

Company Information



- Displays essential details such as company name, physical address, phone number, and email address.
- May also contain a **link to the About Us page** or any other relevant company information.

Quick Links



- Provides **direct access** to commonly accessed pages such as:
  - Policies
  - FAQ
  - Careers
  - Sitemap
  - Partners
  - Contact Us

Social Media Icons and Links

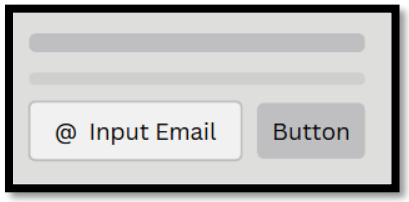


- Allows users to connect with the brand on various platforms, such as:
  - Facebook
  - Instagram

- Twitter
- LinkedIn

- The footer includes an embedded Facebook fanpage, allowing users to:
- View recent posts.
- Follow Maverick Dresses directly from the site.
- Engage with brand updates without leaving the page.

Newsletter Signup



- Includes an email input field where users can subscribe to newsletters and promotional offers.

Payment Methods



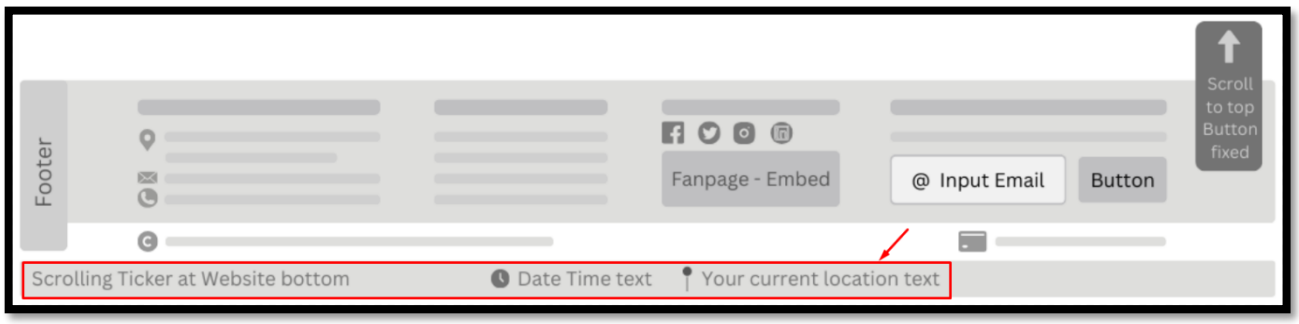
- Displays **supported payment options** for transparency, such as:
- Visa
- MasterCard
- PayPal
- …

Copyright Notice



- Located at the **very bottom of the footer**, displaying ownership rights and legal disclaimers.

Scrolling Ticker Section



- A **persistent ticker located at the very bottom of the page**.
- Make sure there is a button that asks the user to give permission for the browser to know the user's current location. In case the user forgets to give permission when first entering the website, the button will be displayed.
- Displays **real-time information**, including:
  - **Current Date & Time**
  - **User's Geolocation (City, Country)** (via **Geolocation API**).

Scroll to Top Button



- Positioned at the **bottom-right corner of the page**.

- A **fixed floating button** allowing users to instantly scroll back to the top of the page.
- Enhances **usability, especially for pages with long content**.

4.3. Page Layout Sections Overview

The following sections define the content structure within different pages, helping organize information effectively and improving navigation.

Hero Section (Banner)

- The **first section users see** on the homepage.
- Contains **large images or sliders** showcasing:
  - Promotions
  - Best-selling products
  - Limited-time discounts
- Includes call-to-action buttons such as *Shop Now* or *Discover More*.

Main Content Section

- Central area of each page containing relevant content.
  - Product pages display product images, descriptions, and filtering options.
  - The About Us page includes the brand's history, values, and mission.
  - Support pages provide company policies, FAQs, and customer assistance information.

Sidebar (Optional, for Some Pages)

- Typically present on **product listing pages or blogs**.
- Includes filtering options such as price, size, category, and availability.

- May also display related promotions or featured content.

Interactive Features Section

- Enhances user experience with elements such as dynamic sliders, accordions for FAQs, and live updating product widgets.

Call-to-Action (CTA) Section

- Encourages user interaction through newsletter signups, add-to-cart buttons, and limited-time offer banners.

4.4. Important Notes

Product Titles & Breadcrumbs:

- If a **product title** is too long, it will be **automatically truncated with "..."** after three lines to **prevent layout distortion**. To view the full title, users must **click on the product** for more details.
- **Breadcrumbs** will also be truncated if they exceed a certain length to **maintain a clean and structured navigation flow**.

4.5. Layout & Responsiveness

The website is **fully responsive**, ensuring **optimal performance across mobile devices, tablets, and all major web browsers** as per the project requirements. The design dynamically adjusts to different screen sizes, providing a **seamless and user-friendly experience** on **Chrome, Firefox, Safari, Edge, and other modern browsers**.

Next Steps

With the **Header, Footer, and Key Layout Sections** defined, the next step is to break down the structure of individual pages in more detail, starting with the **Homepage Layout**.

Testing document

1. Authentication and Access - User Input Validation

No.	Feature Tested	Remarks
1	**Username or email**	Required; trim whitespace; reject an unknown or disabled account.
2	**Password**	Required; invalid credentials return a safe error without revealing account state.
3	**Role and token**	JWT, expiry, and endpoint role are checked before protected access.
4	**Refresh session**	Refresh is accepted only for a valid session

		and is rejected after account disablement.

2. Organization and Account - User Input Validation

No.	Feature Tested	Remarks
1	**Department name**	Required; trim whitespace; reject duplicate names within the organization.
2	**Staff and role**	Required staff identifier and an allowed role; role changes are traceable.
3	**Account status**	Only valid active/inactive transitions are accepted.
4	**Search and pagination**	Filters are bounded and

		lists use the agreed default and maximum page size.

3. Patient - User Input Validation

No.	Feature Tested	Remarks
	Patient identifiers	Required fields are validated; duplicate patient identifiers are rejected.
	Contact and demographic data	Formats and permitted lengths are checked before save.
	Search result	Only permitted patient fields are returned to the

		requesting role.

4. Appointment - User Input Validation

No.	Feature Tested	Remarks
1	**Patient and clinician**	Both identifiers are required and must exist in the approved context.
2	**Date and time**	Required format; invalid or conflicting appointments are rejected.
3	**Status transition**	Only an approved lifecycle transition is accepted.
4	**Cancel/check-in**	Action is role checked and the resulting state is visible to relevant users.

5. Clinical Record and Diagnosis - User Input Validation

No.	Feature Tested	Remarks
	Clinical episode	Appointment and patient linkage is required before a record is saved.
	Required clinical fields	Validation covers the fields needed for the current record type.
	Diagnosis	Diagnosis is linked to the correct episode and

		authorized clinician.

6. Laboratory - User Input Validation

No.	Feature Tested	Remarks
1	**Laboratory request**	Test type, patient, episode, and requester are required.
2	**Processing status**	Only valid request-to-processing transitions are accepted.
3	**Result values**	Values and required conclusion fields are validated before completion.
4	**Result publication**	Author, time, permissions, and notification behavior are checked.

7. Pharmacy and Inventory - User Input Validation

No.	Feature Tested	Remarks
1	**Medicine and quantity**	Required; quantity must be positive and within available stock.
2	**Prescription eligibility**	Patient, prescription, payment, and status are checked before dispensing.
3	**Concurrent stock**	No negative stock and no duplicate issue under

		retry or concurrent requests.
4	**Failure event**	Insufficient stock and processing failure return a traceable outcome.

8. Billing, Payment and Dispensing - User Input Validation

No.	Feature Tested	Remarks
1	**Invoice**	Only approved

		charges and a valid open invoice can be paid.
2	**Payment amount**	Amount and method are required and validated.
3	**Saga correlation**	Payment and dispensing share a correlation ID and explicit states.
4	**Compensation**	Downstream failure produces compensation or a manual-review state.

9. Notification and Reporting - User Input Validation

No.	Feature Tested	Remarks

1	**Recipient**	Notification is addressed only to the permitted owner.
2	**Duplicate event**	Repeated event does not create a duplicate business notification.
3	**Message retrieval**	User sees only messages allowed by role and ownership.
4	**Report filters**	Date and organization filters return explainable approved totals.

10. Inpatient and Surgery - User Input Validation

No.	Feature Tested	Remarks
	Admission or case	Patient and responsible clinician are required.
	Bed or schedule	Conflicts and invalid transitions are rejected.
	Role and ownership	Extension actions follow the same role and service ownership rules.
	Outcome trace	State, actor, timestamp, and correlation information remain available.

11. Web/Mobile Client - User Input Validation

No.	Feature Tested	Remarks
1	**Client session**	Unauthenticated access is redirected or rejected according to route policy.
2	**Loading and empty states**	The UI communicates progress and no-result conditions clearly.
3	**Error handling**	Validation and server errors are actionable without exposing

		sensitive details.

Final checklist

No.	Review 01 aspect tested	Suggestion / Remarks
1	Project scope covers organization, patients, appointments, clinical records, laboratory, pharmacy, billing, notifications, and reporting.	Review 01 evidence prepared / To be checked
2	The team roster, supervisor, class, and submission date are confirmed in the project records.	Review 01 evidence prepared / To be checked
3	The problem statement identifies duplicated data, unclear status, manual handover, stock risk, saga risk, and limited visibility.	Review 01 evidence prepared / To be checked
4	Actors and roles are documented: ADMIN, MANAGER, DOCTOR, NURSE, LAB_TECH, PHARMACIST, CASHIER, PATIENT, and SYSTEM.	Review 01 evidence prepared /

	Prepared by (Student)	Approved by (Faculty)
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

		To be checked
5	The solution identifies the API Gateway, service discovery, bounded services, PostgreSQL ownership, and RabbitMQ messaging.	Review 01 evidence prepared / To be checked
6	Authentication and authorization requirements are written with invalid and unauthorized cases.	Review 01 evidence prepared / To be checked
7	Patient identity and duplicate prevention requirements are written.	Review 01 evidence prepared / To be checked
8	Appointment lifecycle and invalid transition requirements are written.	Review 01 evidence prepared / To be checked

9	Clinical record, diagnosis, laboratory, and result traceability requirements are written.	Review 01 evidence prepared / To be checked
10	Pharmacy stock, concurrency, idempotency, and dispensing requirements are written.	Review 01 evidence prepared / To be checked
11	Billing, payment, dispensing, saga, and compensation requirements are written.	Review 01 evidence prepared / To be checked
12	Notification ownership and duplicate-event behavior are written.	Review 01 evidence prepared / To be checked
13	Reporting dimensions and explainable totals are written.	Review 01 evidence prepared /

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

		To be checked
14	Hardware and software requirements identify Java 21, Maven, PostgreSQL, RabbitMQ, Next.js, React, TypeScript, and pnpm.	Review 01 evidence prepared / To be checked
15	Non-functional baselines cover security, data ownership, performance, resilience, consistency, observability, testing, and usability.	Review 01 evidence prepared / To be checked
16	The Review 01 boundary excludes insurance, PACS/DICOM, laboratory devices, production bank integration, and multi-region recovery.	Review 01 evidence prepared / To be checked
17	Repository alignment is recorded for backend contexts, gateway, discovery, Web pages, tests, Mobile readiness, and documentation.	Review 01 evidence prepared / To be checked

18	Review 01 validation tables are completed for the core user inputs and failure paths.	Review 01 evidence prepared / To be checked
19	The team records remaining approval items for production SLA, RTO/RPO, log retention, and accessibility target.	Review 01 evidence prepared / To be checked

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

Tasksheet

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

No	Member	Contents	Page Name	Activity Plan	Date of Preparation of Activity Plan		
				Planed Start Date	Actual Start Date	Actual Days	Status
1	Vinh	Appointment list, filters, detail, create, edit, arrival, cancellation; role and state guards.	/appointments /appointments/new /appointments/[appointmentId] /appointments/[appointmentId]/edit	2026-07-28	2026-07-28	7	DONE
2	Vinh	Medical-record lookup, detail, create, symptom update, diagnosis entry, disposition, and immutable completed-state presentation.	/records /records/new /records/[recordId] /records/[recordId]/edit	2026-07-28	2026-07-28	7	DONE

	Prepared by (Student)	Approved by (Faculty)
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

3	Vinh	Laboratory queue, filters, detail, request creation, specimen processing, result entry, and cancellation.	/lab /lab/new /lab/[testId]	2026-07-28	2026-07-28	8	DONE
4	Vinh	Inpatient admission list/detail, admission creation, and bed list. Only admission-create command slice delivered.	/inpatient /inpatient/new /inpatient/[admissionId] /inpatient/beds	2026-10-08	2026-10-08	2	IN PROGRESS
5	Vinh	SHARED: Clinical, Laboratory, and Inpatient API/component tests covering permissions, validation, terminal states, retry, and correlation IDs.	Planned: /appointments, /records, /lab, /inpatient test suites	2026-10-12	—	—	PLANNED

6	Vinh	Complete Inpatient bed assignment, transfer, and release after verifying roles, DTOs, transitions, and conflicts.	Planned: /inpatient/[admissionId]/bed	2026-10-15	—	—	VERIFY CONTRACT
7	Vinh	Add admission and pre-admission cancellation controls after verifying bed placement, financial clearance, permissions, and Gateway routing.	Planned: /inpatient/[admissionId]/admission-state	2026-10-17	—	—	VERIFY CONTRACT
8	Vinh	Expose Clinical check-in, examination start, record completion, and admission-referral commands after backend/Gateway activation.	Planned: controls in /appointments/[appointmentId], /records/[recordId]	2026-10-19	—	—	BLOCKED

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

9	Vinh	Implement Clinical-to-Laboratory test-request handoff using test code and producer-generated order ID.	Planned: /records/[recordId]/lab-orders	2026-10-22	—	—	BLOCKED
10	Vinh	Add Inpatient treatment entries, immutable corrections, order references, bed history, discharge, and administrative-close presentation.	Planned: /inpatient/[admissionId]/care, /inpatient/[admissionId]/order-references	2026-10-26	—	—	VERIFY CONTRACT
11	Vinh	Add bed creation/update controls for administrators and managers; validate bed DTOs, selectors, uniqueness, and occupied-bed restrictions.	Planned: /inpatient/beds/admin	2026-10-28	—	—	VERIFY CONTRACT

12	Vinh	Integrate Inpatient deposit, top-up, settlement, Pharmacy, Surgery, Report, and Notification timeline data after approved handoffs.	Planned: integrated timeline in /inpatient/[admissionId]	2026-10-29	—	—	BLOCKED
13	Huy	Pharmacy drug catalogue/detail/create, stock adjustment, prescription creation/lookup/detail, dispense/cancel, and outbox replay.	/pharmacy /pharmacy/drugs /pharmacy/prescriptions /pharmacy/admin/outbox	2026-07-28	2026-07-28	5	DONE

14	Huy	Daily metrics, monthly revenue, daily detail, and top-dispensed medicine reports; validation, zero states, errors, and retry.	/reports	2026-07-28	2026-07-28	3	DONE
15	Huy	SHARED: Pharmacy validation, permission, terminal-state, API-error, and replay tests.	Planned: /pharmacy test suite	2026-10-12	—	—	PLANNED
16	Huy	SHARED: Report formatter, filter, permission, zero-value, error, and retry tests.	Planned: /reports test suite	2026-10-14	—	—	PLANNED

17	Huy	Add Inpatient/Surgery operational snapshots, financial views, and capability indicators after verifying controllers, DTOs, roles, and Gateway routing.	Planned: /reports/operations, /reports/financial	2026-10-16	—	—	VERIFY CONTRACT
18	Huy	Add financial rebuild, progress, publication/rollback, source-archive export/restore after verifying commands, limits, and authorization.	Planned: /reports/financial-operations	2026-10-19	—	—	VERIFY CONTRACT
19	Huy	Create Surgery case list and detail workspace after backend routes, authority, DTOs, roles, and Gateway routing are approved.	Planned: /surgery, /surgery/[caseId]	2026-10-22	—	—	BLOCKED

20	Huy	Add Surgery request, readiness, clearance, schedule, execution, recovery, outcome, cancellation, and charge-handoff controls.	Planned: /surgery/[caseId]/workflow	2026-10-26	—	—	BLOCKED
21	Huy	Add Surgery permission, lifecycle, conflict, retry, and integration tests after workspace and fixtures are available.	Planned: Surgery test suite	2026-10-29	—	—	BLOCKED

22	Hoàng Anh	Organization department/staff list, detail, create, update, staff transfer, and account administration.	/organization /organization/departments/new /organization/departments/[departmentId] /organization/departments/[departmentId]/edit /organization/staff/new /organization/staff/[staffId] /organization/staff/[staffId]/edit /organization/accounts	2026-07-28	2026-07-28	3	DONE
23	Hoàng Anh	Patient paginated search, detail, create, update, delete, Vietnamese wire mapping, and immutable identity-number behavior.	/patients /patients/new /patients/[patientId] /patients/[patientId]/edit	2026-07-28	2026-07-28	6	DONE

24	Hoàng Anh	SHARED: Organization partial-failure, pagination, empty-state, role-gate, account, and retry tests.	Planned: /organization test suite	2026-10-12	—	—	PLANNED
25	Hoàng Anh	SHARED: Patient contract, validation, permission, deletion, empty/error, and retry tests.	Planned: /patients test suite	2026-10-14	—	—	PLANNED
26	Hoàng Anh	Provide authorized patient, doctor, and department lookup contracts for Clinical, Laboratory, Billing, and Report selectors.	Planned: shared forms in /appointments, /lab, /billing, /reports	2026-10-16	—	—	VERIFY CONTRACT

27	Hoàng Anh	Integrate stable authenticated staffId, patient identity, and account-to-subject mapping across Gateway and Organization.	Planned: /login and session integration	2026-10-19	—	—	VERIFY CONTRACT
28	Hoàng Anh	Add secure patient self-service profile after Gateway identity and Patient ownership contracts are approved.	Planned: /patients/me	2026-10-22	—	—	BLOCKED
29	Hoàng Anh	Provide identity handoff for patient-owned appointments, results, prescriptions, invoices, receipts, payments, and notifications.	Planned: identity contract for patient portal routes	2026-10-26	—	—	BLOCKED

30	Lộc	Billing patient invoice lookup, detail, creation from server-owned fees, payment state, and fee breakdown.	/billing /billing/new /billing/[invoiceId]	2026-07-28	2026-07-28	3	DONE
31	Lộc	Notification patient history, privacy-safe secured detail, and administrator manual-send workflow.	/notifications /notifications/[notificationId] /notifications/send	2026-07-28	2026-07-28	3	DONE
32	Lộc	SHARED: Billing pagination, monetary display, permission, terminal-state, API-error, retry, and payment-flow tests.	Planned: /billing test suite	2026-10-12	—	—	PLANNED

33	Lộc	SHARED: Notification pagination, channel/status, privacy, cross-patient denial, error, and retry tests.	Planned: /notifications test suite	2026-10-14	—	—	PLANNED
34	Lộc	Add ledger payment/refund, deposit allocation, settlement preview, reconciliation, settlement, and final-settlement controls.	Planned: /billing/accounts/[accountId], /billing/transactions/[transactionId]	2026-10-16	—	—	VERIFY CONTRACT
35	Lộc	Add administrator/manager revenue filtering by department/date after validating RevenueByDeptDTO and Gateway routing.	Planned: /billing/revenue	2026-10-17	—	—	VERIFY CONTRACT

	Prepared by (Student)	**Approved by (Faculty)**
	Project Group No: 02	LÊ THANH NHÂN
Signature		
Date		

36	Lộc	Add prescription-adjustment and Surgery-cancellation refund-due read models, without exposing refund commands.	Planned: /billing/prescription-adjustments/[adjustmentId]/refunds-due, /billing/surgery-cancellations/[cancellationId]/refunds-due	2026-10-18	—	—	VERIFY CONTRACT
37	Lộc	Add Billing administrator outbox replay after verifying event discovery, maintenance settings, correlation IDs, and Gateway routing.	Planned: /billing/admin/outbox	2026-10-20	—	—	VERIFY CONTRACT
38	Lộc	Add Notification read-state and authorized retry commands after endpoint, role, lifecycle, and privacy contracts are available.	Planned: controls in /notifications/[notificationId]	2026-10-19	—	—	BLOCKED

39	Lộc	Add authenticated My Notifications after Gateway/Patient identity mapping and backend ownership checks are enforced.	Planned: /notifications/me	2026-10-22	—	—	BLOCKED
40	Lộc	Define Billing currency semantics in coordination with Report; frontend must not infer currency from locale.	Planned: /billing and shared /reports display contract	2026-10-26	—	—	VERIFY CONTRACT

41	Vinh (Shared Coordinator)	SHARED: Run authenticated browser smoke tests across all delivered dashboard contexts, covering roles, privacy, correlation IDs, and critical workflows.	Planned: all delivered dashboard routes	2026-11-02	—	—	PLANNED

Acknowledgments and Declaration

Project Documentation for Maverick Dresses

Completed on 18/02/2025

The documentation for the **Maverick Dresse**s project was completed on February 18, 2025. Our team would like to express our gratitude to our supervisor, **Mr. LÊ THANH NHÂN**, for his guidance and valuable feedback throughout the project. We also appreciate the constructive input from our fellow students, which contributed to the refinement of our work.

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This document is the result of our team's efforts, based on the knowledge we have gained during our studies and additional research. We confirm that the content is original and accurate, with no unauthorized copying or plagiarism. Our team is committed to maintaining academic integrity and ensuring the quality of this project.

This is a complete document that reflects the efforts and research of our team as students who are still learning and have not yet entered the professional workforce. We are grateful to our teacher for his support throughout this project. The content is based on the knowledge we have acquired during our studies and reliable sources.